

Scenario 10: Accessing Ryu Web GUI from Host Machine

Question

A developer is running Mininet in a VM using the default "NAT" network setting. They start a Ryu controller with a web GUI on port 8080 inside the VM. They try to access the GUI from their host machine's browser using the VM's IP address but fail. What is the cause of this problem, and what is the recommended solution?

Answer

The problem is caused by the **NAT (Network Address Translation) network mode** used by the virtual machine. In NAT mode, the VM is placed behind a virtual NAT network. The VM can generally access the host/network and the Internet, but services running inside the VM are not automatically reachable from the host using the VM's NAT-side IP address.

Cause of the Problem

- 1. VM is behind NAT:** The virtualization software creates a private virtual network for the VM. The VM's IP address belongs to this private NAT network rather than behaving like a normal machine directly connected to the host's LAN.
- 2. Port 8080 is inside the VM:** Ryu's web GUI is listening on port 8080 within the VM. NAT does not automatically expose that service to the host machine.
- 3. Host cannot directly reach the service:** Entering the VM's NAT IP and port 8080 from Chrome may fail because there is no appropriate NAT port-forwarding rule exposing the Ryu service to the host.

Recommended Solution

The recommended solution is to change the VM's network adapter from **NAT** to **Host-Only Adapter** (or another suitable host-reachable network mode such as Bridged Adapter, depending on the setup). Host-Only networking gives the host and VM addresses on the same private virtual network, allowing the host browser to reach services running inside the VM.

Steps to Fix the Issue

1. Shut down the Mininet/Ryu virtual machine if required by the virtualization software.
2. Open the VM's network settings.
3. Change the network adapter from **NAT** to **Host-Only Adapter**.
4. Start the VM again and find its new IP address using a command such as **ip addr** or **ifconfig**.
5. Make sure the Ryu controller is running and its web server is listening on port **8080**.
6. From the host machine's browser, open **http://<VM-IP>:8080**.

Alternative Solution: NAT Port Forwarding

If NAT must be retained, configure a **port-forwarding rule** in the virtualization software. For example, a host port can be forwarded to port 8080 of the VM. The browser can then access the Ryu GUI through the forwarded host port. However, for a simple development/lab environment where the host needs to access services inside the VM, Host-Only networking is usually the simpler recommended approach.

NAT vs. Host-Only

Feature	NAT	Host-Only
VM network	Behind virtual NAT	Private network shared with host
Host → VM access	Not automatically available	Normally available
Ryu GUI access	Needs port forwarding	Directly via VM IP
Recommended for this case	Only if port forwarding is configured	Yes

Conclusion

The failure is primarily a **network connectivity/NAT configuration issue**, not a Mininet or Ryu GUI problem. Because the VM is using NAT, the Ryu service on port 8080 is not directly exposed to the host. The recommended fix is to use a **Host-Only Adapter** so the host can directly communicate with the VM, or configure **NAT port forwarding** if NAT must be retained.

Short Exam Answer

The issue occurs because the VM is using NAT, which places it behind a virtual NAT and does not automatically expose the Ryu web service on port 8080 to the host. The recommended solution is to change the VM network to **Host-Only Adapter** and access the GUI using the VM's host-only IP and port 8080. Alternatively, configure NAT port forwarding from a host port to VM port 8080.